# Vincent T. Cook

**MANUFACTURING & PRODUCTION**

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Aerospace engineering senior with CAD assembly design, motor test-bench documentation, circuitry design, and battery testing experience. Apply structures, materials, wind-tunnel testing, and project planning to production engineering.

## EDUCATION

San Jose State University | San Jose, CA
B.S. Aerospace Engineering | Expected May 2027 | Enrolled Sep 2022

## TECHNICAL SKILLS

Design & documentation: SolidWorks, Autodesk Inventor, 3D CAD, assembly instructions, stress analysis
Analysis & tools: simulation and modeling, CFD, thermal testing, MATLAB, Python, C++, Java

## ENGINEERING EXPERIENCE & PROJECTS

**SAE Spartan Racing**
EV Powertrain Intern | Aug 2023–May 2025
• Developed 3D CAD and assembly instructions for an electric motor test bench; integrated mechanical/electrical components and aligned engineering/manufacturing practices with teammates.

**Hybrid Electric / Hydrogen Fuel-Cell Aircraft | SJSU Senior Project**
Aerospace Engineering Team Member — In Progress | 2026–Present
• Contribute wing-model designs and planning for a hybrid battery-electric/hydrogen fuel-cell model aircraft.
• Planned: CAD, CFD/thermodynamics, construction/testing, battery/fuel-system electronics, and commercial-scale modeling; team flight target spring 2027.

**SJSU Robotics**
Mechanical Designer | Aug 2024–Present
• Designed a transmitter mast with camera and sensor mounts; used stress analysis to reinforce high-load regions and select lighter materials for lower-load sections.

**BROADSWORD | Reusable Rocket Concept**
CAD & Programming Subteam Lead | Sep–Dec 2023
• Created landing-leg CAD and code producing trajectory-angle-versus-time plots for a reusable rocket concept; delegated CAD/programming tasks, coordinated subteam work, and provided technical feedback.

**NASA L'SPACE Mission Concept Academy**
Mechanical Engineering Team Member — Educational Program | Fall 2026–Present
• Help plan science approaches and evaluate mechanical components for a robotic lunar surface payload mission concept.

**FIRST Robotics Team 3925**
Electrical Lead — High School Robotics | 2020–2022
• Designed circuitry and tested batteries for the team's robot.

## SELECTED COURSEWORK

• AE 20 | Created high-level motor designs in introductory CAD coursework.
• AE 112 / AE 114 | Studied structural dynamics and physical/material properties.
• AE 160 / AE 162 | Used open wind-tunnel lab simulations to test wing airfoils and other objects.

## WORK EXPERIENCE

**PayPal Stadium | Guest Representative | 2026 Soccer Season**
• Worked San Jose Earthquakes and Bay FC women's team matches.

**The Tech Interactive | Educator I | 2024–2025**
• Led STEM labs/design challenges and explained procedures and technical concepts.

**Self-Employed | Private Swim Instructor | 2023**
• Scheduled clients and tailored swim lessons to individual needs.

**YMCA & Ventura Aquatics Center | Swim Instructor / Coach / Lifeguard | 2021–2024**
• Balanced coaching/lifeguard work and adapted swim lessons to varied abilities.